Lab Questions – 21st Nov 2025

PART 1 — SQL Queries

Q1. Create a table named students with fields:

- stdid INT PRIMARY KEY
- stdname VARCHAR(50)
- age INT
- city VARCHAR(50)

Solution:

```
mysql> create table students(stdid INT PRIMARY KEY,stdname VARCHAR(50),age INT, city VARCHAR(50));
Query OK, 0 rows affected (0.06 sec)

mysql> desc students;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| stdid   | int         | NO   | PRI | NULL    |       |
| stdname | varchar(50) | YES  |     | NULL    |       |
| age     | int         | YES  |     | NULL    |       |
| city    | varchar(50) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)
```

Q2. Insert the following records into the students table:

stdid	stdname	age	city
1	Rohan	20	Pune
2	Meera	22	Mumbai
3	Arjun	21	Delhi
4	Kavya	23	Pune
5	Neha	22	Kolkata

Solution:

```
mysql> INSERT INTO students (stdid, stdname, age, city) VALUES
    -> (1, 'Rohan', 20, 'Pune'),
    -> (2, 'Meera', 22, 'Mumbai'),
    -> (3, 'Arjun', 21, 'Delhi'),
    -> (4, 'Kavya', 23, 'Pune'),
    -> (5, 'Neha', 22, 'Kolkata');
Query OK, 5 rows affected (0.08 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

Q3. Display all student records.

Solution:

```
mysql> select * from students;
+-------+---------+------+---------+
| stdid | stdname | age  | city    |
+-------+---------+------+---------+
|     1 | Rohan   |   20 | Pune    |
|     2 | Meera   |   22 | Mumbai  |
|     3 | Arjun   |   21 | Delhi   |
|     4 | Kavya   |   23 | Pune    |
|     5 | Neha    |   22 | Kolkata |
+-------+---------+------+---------+
5 rows in set (0.00 sec)
```

Q4. Display only the name and age of all students.

Solution:

```
mysql> select stdname, age from students;
+---------+------+
| stdname | age  |
+---------+------+
| Rohan   |   20 |
| Meera   |   22 |
| Arjun   |   21 |
| Kavya   |   23 |
| Neha    |   22 |
+---------+------+
5 rows in set (0.00 sec)
```

Q5. Display students who are from Pune.

Solution:

```
mysql> select * from students where city = 'Pune';
+-------+---------+------+------+
| stdid | stdname | age  | city |
+-------+---------+------+------+
|     1 | Rohan   |   20 | Pune |
|     4 | Kavya   |   23 | Pune |
+-------+---------+------+------+
2 rows in set (0.00 sec)
```

Q6. Display students whose age is greater than 21.

Solution:

```
mysql> select * from students where age > 21;
+-------+---------+------+---------+
| stdid | stdname | age  | city    |
+-------+---------+------+---------+
|     2 | Meera   |   22 | Mumbai  |
|     4 | Kavya   |   23 | Pune    |
|     5 | Neha    |   22 | Kolkata |
+-------+---------+------+---------+
3 rows in set (0.00 sec)
```

Q7. Display students in descending order of age.

Solution:

```
mysql> select * from students order by age desc;
+-------+---------+------+---------+
| stdid | stdname | age  | city    |
+-------+---------+------+---------+
|     4 | Kavya   |   23 | Pune    |
|     2 | Meera   |   22 | Mumbai  |
|     5 | Neha    |   22 | Kolkata |
|     3 | Arjun   |   21 | Delhi   |
|     1 | Rohan   |   20 | Pune    |
+-------+---------+------+---------+
5 rows in set (0.00 sec)
```

Q8. Count how many students belong to each city. (Use GROUP BY).

Solution:

```
mysql> select city, count(city) as total_students from students group by city;
+---------+----------------+
| city    | total_students |
+---------+----------------+
| Pune    |              2 |
| Mumbai  |              1 |
| Delhi   |              1 |
| Kolkata |              1 |
+---------+----------------+
4 rows in set (0.00 sec)
```

Q9. Display students whose name starts with ‘K’. (Use LIKE).

Solution:

```
mysql> select * from students where stdname like 'K%';
+-------+---------+------+------+
| stdid | stdname | age  | city |
+-------+---------+------+------+
|     4 | Kavya   |   23 | Pune |
+-------+---------+------+------+
1 row in set (0.00 sec)
```

Q10. Delete student whose stdid = 5.

Solution:

```
mysql> delete from students where stdid = 5;
Query OK, 1 row affected (0.03 sec)

mysql> select * from students;
+-------+---------+------+--------+
| stdid | stdname | age  | city   |
+-------+---------+------+--------+
|     1 | Rohan   |   20 | Pune   |
|     2 | Meera   |   22 | Mumbai |
|     3 | Arjun   |   21 | Delhi  |
|     4 | Kavya   |   23 | Pune   |
+-------+---------+------+--------+
4 rows in set (0.00 sec)
```

PART 2 — ALTER COMMAND QUESTIONS

Q11. Add a new column contact VARCHAR(15) to the students table.

Solution:

```
mysql> ALTER table students Add contact varchar(15);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc students;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| stdid   | int         | NO   | PRI | NULL    |       |
| stdname | varchar(50) | YES  |     | NULL    |       |
| age     | int         | YES  |     | NULL    |       |
| city    | varchar(50) | YES  |     | NULL    |       |
| contact | varchar(15) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

Q12. Modify the data type of city column to VARCHAR(100).

Solution:

```
mysql> ALTER table students MODIFY city varchar(100);
Query OK, 4 rows affected (0.14 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> desc students;
+---------+--------------+------+-----+---------+-------+
| Field   | Type         | Null | Key | Default | Extra |
+---------+--------------+------+-----+---------+-------+
| stdid   | int          | NO   | PRI | NULL    |       |
| stdname | varchar(50)  | YES  |     | NULL    |       |
| age     | int          | YES  |     | NULL    |       |
| city    | varchar(100) | YES  |     | NULL    |       |
| contact | varchar(15)  | YES  |     | NULL    |       |
+---------+--------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

Q13. Rename the column stdname to student_name.

Solution:

```
mysql> ALTER table students CHANGE stdname student_name VARCHAR(50);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Q14. Drop the column contact from the table.

Solution:

```
mysql> ALTER table students drop column contact;
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Q15. Add a new column gender ENUM('M','F').

Solution:

```
mysql> ALTER TABLE students
    -> ADD COLUMN gender ENUM('M','F');
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc students;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| stdid        | int          | NO   | PRI | NULL    |       |
| student_name | varchar(50)  | YES  |     | NULL    |       |
| age          | int          | YES  |     | NULL    |       |
| city         | varchar(100) | YES  |     | NULL    |       |
| gender       | enum('M','F') | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

PART 3 — JOIN PRACTICE

TABLES :

Table: students

stdid	student_name	city
1	Rohan	Pune
2	Meera	Mumbai
3	Arjun	Delhi
4	Kavya	Pune

Table: marks

stdid	subject	marks
1	Maths	88
2	Maths	76
3	Maths	92
5	Maths	67

INNER JOIN

Q16. Display student name and marks of only those students who have matching IDs in both tables. (Students without marks should not appear.)

Solution:

```
mysql>  select student_name, marks from students s inner join marks m on s.stdid = m.stdid;
+--------------+-------+
| student_name | marks |
+--------------+-------+
| Rohan        |    88 |
| Meera        |    76 |
| Arjun        |    92 |
+--------------+-------+
3 rows in set (0.00 sec)
```

LEFT JOIN

Q17. Display all students and their marks. (If marks not available, show NULL.)

Solution:

```
mysql>  select s.stdid, student_name, city, subject, marks from students s left join marks m on s.stdid = m.stdid;
+-------+--------------+--------+---------+-------+
| stdid | student_name | city   | subject | marks |
+-------+--------------+--------+---------+-------+
|     1 | Rohan        | Pune   | Maths   |    88 |
|     2 | Meera        | Mumbai | Maths   |    76 |
|     3 | Arjun        | Delhi  | Maths   |    92 |
|     4 | Kavya        | Pune   | NULL    |  NULL |
+-------+--------------+--------+---------+-------+
4 rows in set (0.02 sec)
```

RIGHT JOIN

Q18. Display all marks records along with student names. (If student doesn’t exist in students table, show NULL.)

Solution:

```
mysql>  select m.stdid, student_name, subject, marks from students s right join marks m on s.stdid = m.stdid;
+-------+--------------+---------+-------+
| stdid | student_name | subject | marks |
+-------+--------------+---------+-------+
|     1 | Rohan        | Maths   |    88 |
|     2 | Meera        | Maths   |    76 |
|     3 | Arjun        | Maths   |    92 |
|     5 | NULL         | Maths   |    67 |
+-------+--------------+---------+-------+
4 rows in set (0.00 sec)
```

CROSS JOIN

Q19. Display all possible combinations of students and subjects. (Use CROSS JOIN between students and marks table to show every pair.) JOIN with Filtering

Solution:

```
mysql>  select s.stdid, student_name, subject, marks from students s cross join marks m;
+-------+--------------+---------+-------+
| stdid | student_name | subject | marks |
+-------+--------------+---------+-------+
|     4 | Kavya        | Maths   |    88 |
|     3 | Arjun        | Maths   |    88 |
|     2 | Meera        | Maths   |    88 |
|     1 | Rohan        | Maths   |    88 |
|     4 | Kavya        | Maths   |    76 |
|     3 | Arjun        | Maths   |    76 |
|     2 | Meera        | Maths   |    76 |
|     1 | Rohan        | Maths   |    76 |
|     4 | Kavya        | Maths   |    92 |
|     3 | Arjun        | Maths   |    92 |
|     2 | Meera        | Maths   |    92 |
|     1 | Rohan        | Maths   |    92 |
|     4 | Kavya        | Maths   |    67 |
|     3 | Arjun        | Maths   |    67 |
|     2 | Meera        | Maths   |    67 |
|     1 | Rohan        | Maths   |    67 |
+-------+--------------+---------+-------+
16 rows in set (0.00 sec)
```

Q20. Using INNER JOIN, display students who scored more than 80.

Solution:

```
mysql>  select s.stdid, student_name, city, subject, marks from students s inner join marks m on s.stdid = m.stdid where
 m.marks > 80;
+-------+--------------+-------+---------+-------+
| stdid | student_name | city  | subject | marks |
+-------+--------------+-------+---------+-------+
|     1 | Rohan        | Pune  | Maths   |    88 |
|     3 | Arjun        | Delhi | Maths   |    92 |
+-------+--------------+-------+---------+-------+
2 rows in set (0.00 sec)
```